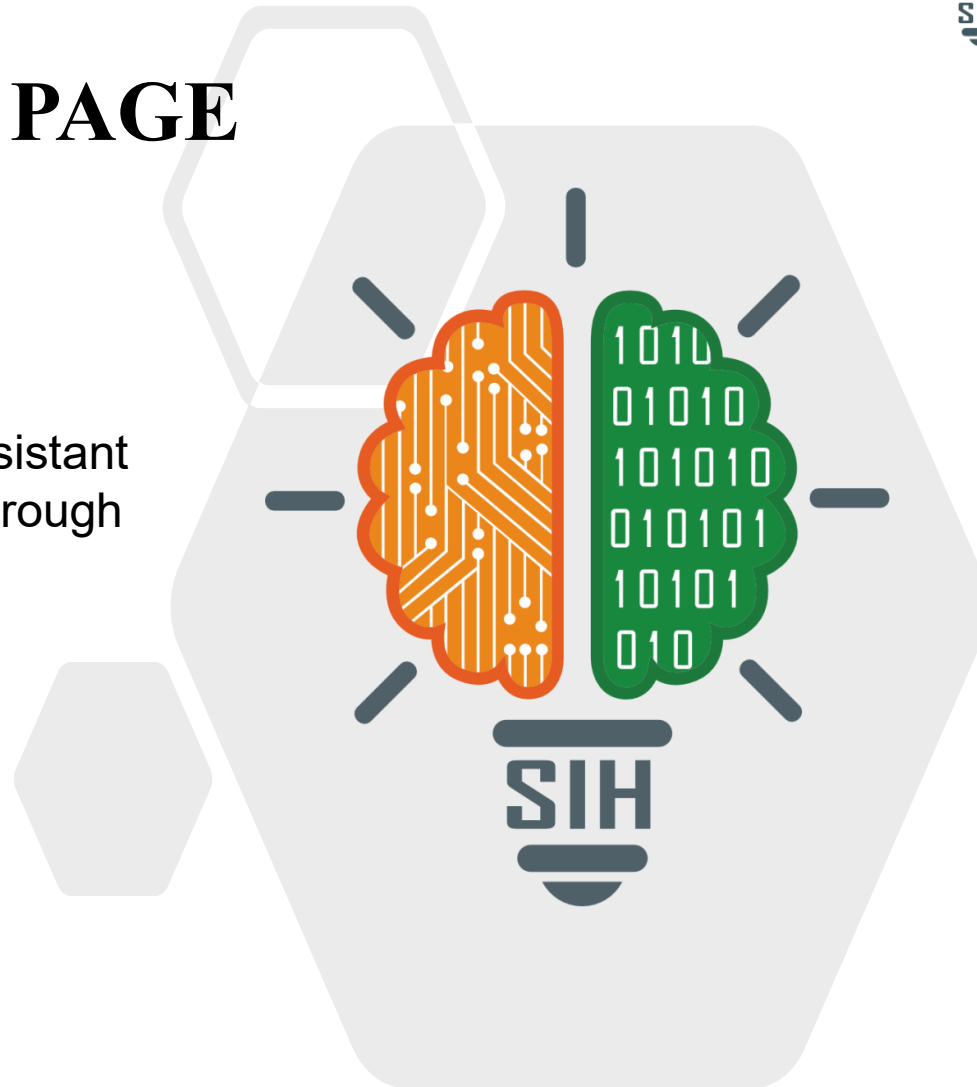


TITLE PAGE

- **Problem Statement ID – 26167**
- **Problem Statement Title –**
SatQuery AI - An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries
- **Theme – Space Technology**
- **PS Category – Software**
- **Team ID –**
- **Team Name – TensorTitans**

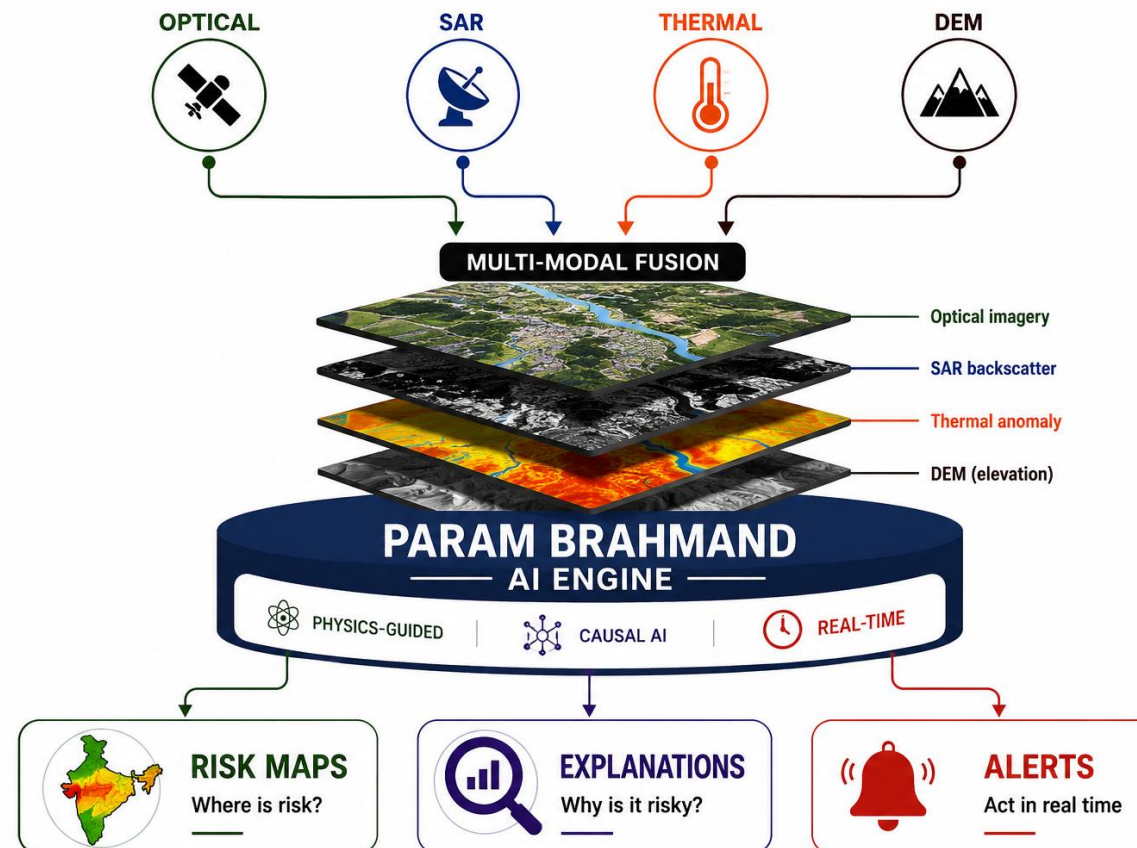


❖ Proposed Solution

An AI-powered geospatial platform that combines Optical, SAR, Thermal and DEM data with Physics-guided and Causal AI to detect environmental risks early and provide explainable, real-time alerts.

Addressing the Problem

-  **Clouds & darkness** → Optical data becomes unreliable
-  **Fragmented datasets** → Hard to build a complete picture
-  **False alarms** → Delayed response & wasted resources
-  **Complex tools** → Difficult for non-expert users



Solution Overview

-  **Multimodal Data**
Optical + SAR + Thermal + DEM
-  **Physics & Causal AI**
Fewer false alarms, better reasoning
-  **Early Risk Prediction**
Probabilistic hazard maps & alerts
-  **Accessible Intelligence**
Simple dashboard + 22 languages

INPUTS



OPTICAL /
MULTISPECTRAL
High Resolution
Imagery



SAR
All-Weather Radar
Data



THERMAL
Heat & Emission
Information

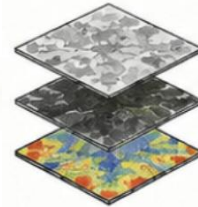


DEM
Elevation & Terrain
Data

1

PHYSICS-FIRST
REPRESENTATION

128-D Physics Manifold
(SIRTI + Indices)

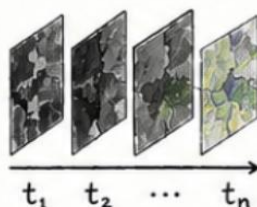


Convert raw data into
physical features
invariant to conditions
& sensors

2

GEO-MAMBA 3.0
BACKBONE

Spatio-Temporal
Understanding



Linear State Space
Model learns and
tracks patterns across
space and time
efficiently.

3

SPECIALIST AGENT
ROUTING (SANKALPA-PARAM)

Navagraha Specialist
Agents



Autonomous router
selects best expert(s)
for the query & task
(e.g. flood,
deforestation, urban)

4

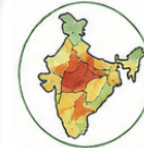
CAUSAL & PHYSICS
VALIDATION

Vivek-Causal + GeoCP-v2
+ Dharma-Chakra



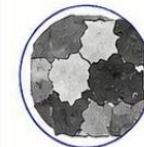
Causal reasoning,
uncertainty quantification
& hard physics rules ensure
reliable and explainable
predictions.

OUTPUTS



RISKS MAPS

Where is the risk?



CHANGE MASKS

What has changed?



ALERTS

Act in real-time



VQA & CAPTIONS

Ask in natural language

Continuous Feedback & Learning

FEASIBILITY AND VIABILITY

FEASIBILITY

(Can we build it?)



ACCESS TO DATA

ISRO satellites (Cartosat, RISAT, Resourcesat) and open datasets (Sentinel, Landsat) available.



PROVEN TECHNOLOGIES

Proven AI & SAR technologies ready to integrate.



COMPUTE & INFRASTRUCTURE

Cloud/HPC + ISRO infrastructure support training & inference.



TECHNICAL INTEGRATION

Modular architecture fuses optical, SAR, DEM & ancillary data.



RISK MITIGATION

Physics-guided models + uncertainty estimation reduce false alarms.

BUILD



VALIDATE



SCALABLE

VIABILITY

(Will it deliver impact?)



STRONG REAL-WORLD NEED

Tackles national challenges: disasters, agriculture, defense, environment.



MEASURABLE IMPACT

Earlier warnings • Better decisions • Fewer losses



SCALABLE & COST EFFECTIVE

Uses existing infrastructure; designed for nationwide scale.



SUSTAINABLE SOLUTION

Supports conservation & climate resilience.



ALIGNED WITH NATIONAL GOALS

Supports Digital India, Atmanirbhar Bharat, disaster-resilient Bharat.

WHO WE IMPACT



DISASTER RESPONDERS (NDMA, SDRF)
All-weather awareness for faster response.



GOVERNMENT & POLICY MAKERS
Better planning, infrastructure & resource decisions.



DEFENCE & STRATEGIC AGENCIES
Real-time monitoring of critical areas & assets.



FARMERS & RURAL COMMUNITIES
Early warnings for crops, water & land health.



RESEARCHERS & STUDENTS
Open insights and datasets for research & innovation.



From Space Data
To Secure Decisions

WHAT WE DELIVER



EARLIER
Risks identified before escalation



SMARTER
Physics + causal reasoning:
Fewer false alarms



LOWER LOSS
Protect infrastructure, crops & resources



SAFER
Better preparedness and response



SCALABLE
From local hazards to national monitoring

PROBLEM & DATASETS

**SIH 2026 – PS 26167 (ISRO SAC)****SatQuery AI**

Source: *SIH 2026 Problem Statement*

**BigEarthNet.txt – 2026**

Large-scale multi-sensor image-text dataset for Earth observation.

Source: <https://txt.bigearth.net>

**BigEarthNet v2.0 / reBEN**

Refined Sentinel-1 + Sentinel-2 dataset for remote sensing benchmarking.

Source: <https://bigearth.net>

EARTH OBSERVATION

**ISRO – Earth Observation Satellites**

Source: <https://www.isro.gov.in>

**ISRO – NISAR Mission**

Source:

<https://www.isro.gov.in/NISARSatellite.html>

**ESA – Copernicus Sentinel Missions**

Source:

https://www.esa.int/Applications/Observing_the_Earth/Copernicus/

RESEARCH METHODS

**PolSAR / PolInSAR****MESMA & Spectral Unmixing****Vision-Language Models (VLMs)****Causal AI – Structural Causal Models****Uncertainty Quantification**

Key Textbook References

James B. Campbell, Randolph H. Wynne & Valerie A. Thomas — *Introduction to Remote Sensing*, 6th Ed.

John A. Richards — *Remote Sensing Digital Image Analysis: An Introduction*, 5th Ed.

Judea Pearl — *Causality: Models, Reasoning, and Inference*, 2nd Ed.